1. PROJECT INITIALIZATION AND PLANNING PHASE

Date	24th Feb 2026
Team ID	259422
Project Title	Online Payments Fraud Detection System
Maximum Marks	3

1.1 Problem Statement

Problem Description

Digital financial transactions are increasingly vulnerable to fraudulent activities. Due to extreme class imbalance and evolving fraud strategies, traditional rule-based systems fail to detect fraud effectively.

Business Impact

- Financial loss
- Customer dissatisfaction
- Brand damage
- Regulatory risk

ML Objective

Develop a supervised classification model to detect fraudulent transactions while minimizing False Negatives.

1.2 Project Proposal (Proposed Solution)

Proposed Approach

Stage	Description
Data Cleaning	Remove irrelevant features
Encoding	Label encoding for categorical variable
Scaling	StandardScaler
Modeling	Decision Tree, Extra Trees & Random Forest
Evaluation	Recall-focused metrics
Deployment	Streamlit Web App

Expected Output

Binary classification:

- 0 → Legitimate
- 1 → Fraud

1.3 Project Planning

Task	Timeline
Data Collection	Week 1
Preprocessing	Week 1
Model Training	Week 2
Model Evaluation	Week 2
Deployment	Week 3